CLASS 01

Two Symbols, Everything Else

How a machine that only knows on and off ended up running the world.

FYS 191MCS26 · Bits, Bytes, and Modern Computing Systems

Mon Sep 14 · Wed Sep 9

HELLO

About me

4 min

Angela Upreti

- What I work on
- How I got here
- Why I wanted to teach this seminar

Ask me about

- Picking classes and a major
- Research on campus
- Anything in the first-year fog

EVERYONE

Icebreaker: two truths and a bit

8 min

Say three things about yourself. One of them is false.

The room votes by holding up fingers:

1st	2nd	3rd
0	1	0

```
1 = I believe you
0 = I don't
```

The point

You just voted in binary.

Three yes/no answers from each person is three bits — and three bits can hold eight different verdicts.

THE ARC

Where this course goes

3 min

A bit on / off → **A byte** 8 bits → **Numbers** and text → **Pictures** and sound → **A computer** all of it

Same two symbols the whole way up.

EVOLUTION

Why only two symbols?

5 min

Gears
Babbage, 1830s

Relays
1930s

Vacuum tubes
ENIAC, 1945

Transistors
1947

Billions of them
your phone

Every one of these is a switch. A switch is either open or closed — so the machine gets exactly two symbols to work with.

GROUP ACTIVITY

15 min

Send a message with nothing but 0s and 1s

Your group must get a message to the group across the room. The only thing you may transmit is a sequence of 0s and 1s.

1. In groups of 3-4, agree on a code — write it down.
2. Encode a short word (3-5 letters) using only 0s and 1s.
3. Swap sheets with another group. Decode theirs.
4. Did it work? If not, where exactly did it break?

YOU NEED

One sheet of paper and a pen per group. No phones.

WATCH FOR

Groups that forget to send the codebook. Groups that can't tell where one letter ends and the next begins. Both failures are the lesson.

What went wrong

4 min

Ambiguity

`1 = A     11 = B     111 = C`

`So what is  1 1 1  ?`

`C?   AAA?   AB?   BA?`

Fix: give every letter the same width.

No shared code

Perfect bits.
No way to read them.

*Fix: agree on the code in advance,
for everyone, forever.*

That agreement is called ASCII. Class 4.

LOGIC GATES

Now: what can you DO with bits?

8 min

NOT
flips it

AND
both

OR
either

XOR
one, not both

Three of these are enough to build arithmetic. All four are enough to build a computer.

REFERENCE

All of it, on one card

2 min

A	B	AND	OR	XOR
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

A	NOT A
0	1
1	0

Look at the XOR column. 1 + 1 = 0, carry the 1. That is addition — and next time we start building with it.

ONE THING TO REMEMBER

Two symbols, agreed on in advance, and rules for combining them.

NEXT TIME

Memory and SSDs — where all these bits actually live.